

Chapter 4: Motion in a Plane

Complete study notes covering scalars, vectors, projectile motion, and circular motion — everything you need to master this chapter.

Provided by ManaranjanStudyS



What We'll Study

Motion in a plane extends one-dimensional kinematics into two dimensions using vectors. This chapter is foundational for understanding projectile motion, relative velocity, and circular motion.

01

Scalars & Vectors

Distinguishing quantities with and without direction

03

Motion in 2D

Position, velocity, and acceleration vectors

02

Vector Operations

Addition, subtraction, resolution, and products

04

Projectile & Circular Motion

Real-world applications of 2D kinematics

Scalars and Vectors

Scalar Quantities

Have **magnitude only** — no direction.

- Mass, temperature, time
- Speed, distance, energy
- Work, power, electric charge

Vector Quantities

Have both **magnitude and direction**.

- Displacement, velocity, acceleration
- Force, momentum, weight
- Electric field, magnetic field

 A vector is represented as $\vec{r}$ or $r\hat{x}$ with an arrow symbol.

Vector Operations

1

Addition (Triangle Law)

Two vectors placed head-to-tail; resultant connects tail of first to head of second. **Commutative:** $A + B = B + A$

2

Parallelogram Law

Resultant $R = \sqrt{A^2 + B^2 + 2AB \cos\theta}$, where θ is the angle between vectors A and B.

3

Resolution of Vectors

Any vector can be split into components: $A_x = A \cos\theta$ and $A_y = A \sin\theta$

4

Scalar (Dot) Product

$A \cdot B = |A||B|\cos\theta$ — gives a scalar. Used to find work done by a force.

Motion in a Plane — Key Concepts

Position Vector

$$\mathbf{r} = x\hat{i} + y\hat{j}$$

Describes the location of a particle in the xy-plane relative to the origin.

Velocity Vector

$$\mathbf{v} = d\mathbf{r}/dt = v_x\hat{i} + v_y\hat{j}$$

Tangent to the path of motion at every point.

Acceleration Vector

$$\mathbf{a} = d\mathbf{v}/dt = a_x\hat{i} + a_y\hat{j}$$

Can change the speed, direction, or both.

- In 2D motion, x and y components are **independent** — treat them separately!

Projectile Motion

A projectile is any object thrown into space that moves under gravity alone. Its path is a **parabola**.

Horizontal Motion

Uniform velocity: $v_x = u \cos\theta$
No acceleration in x-direction

Vertical Motion

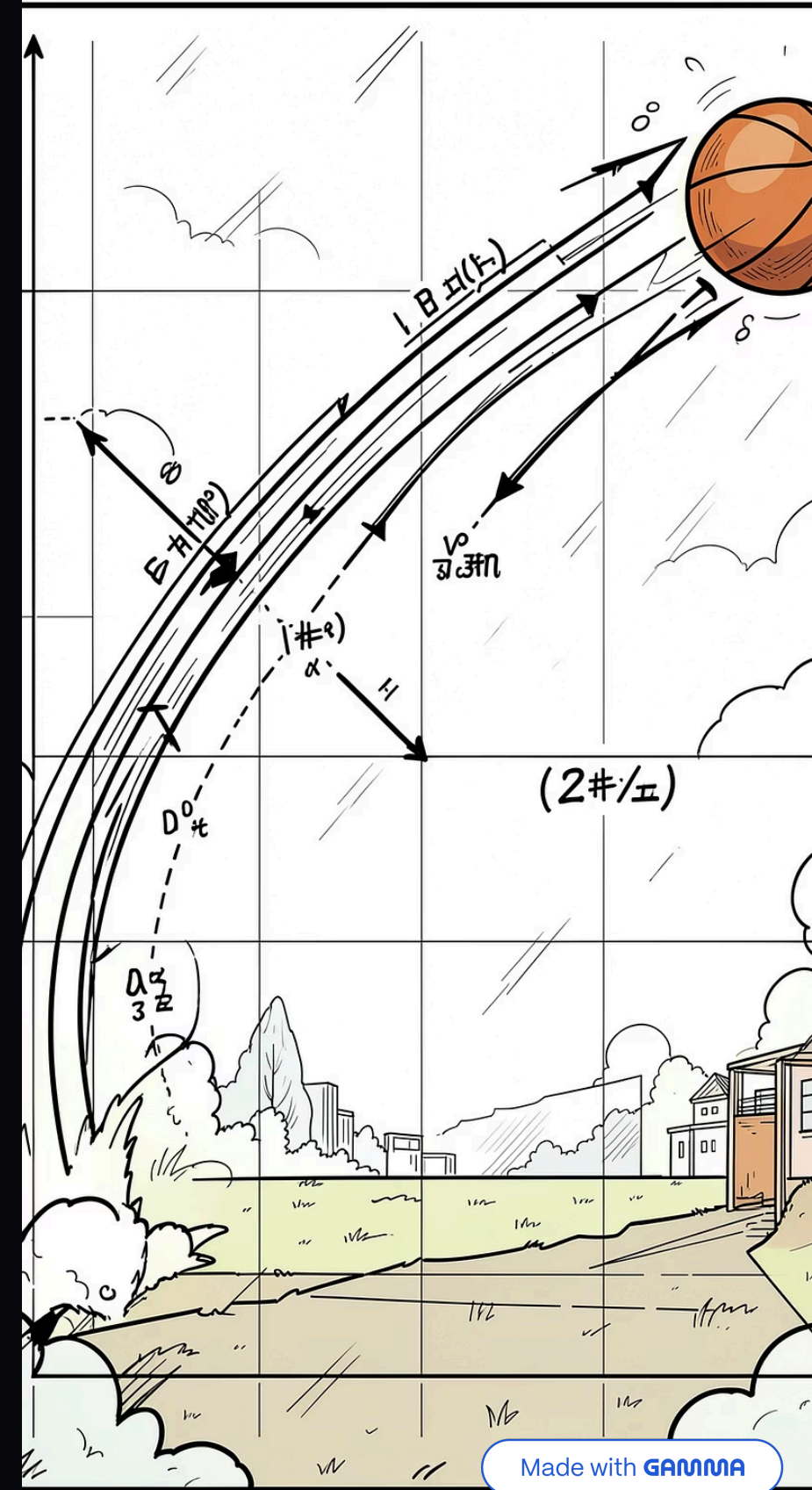
Uniform acceleration: $a_y = -g$
 $v_y = u \sin\theta - gt$

Time of Flight

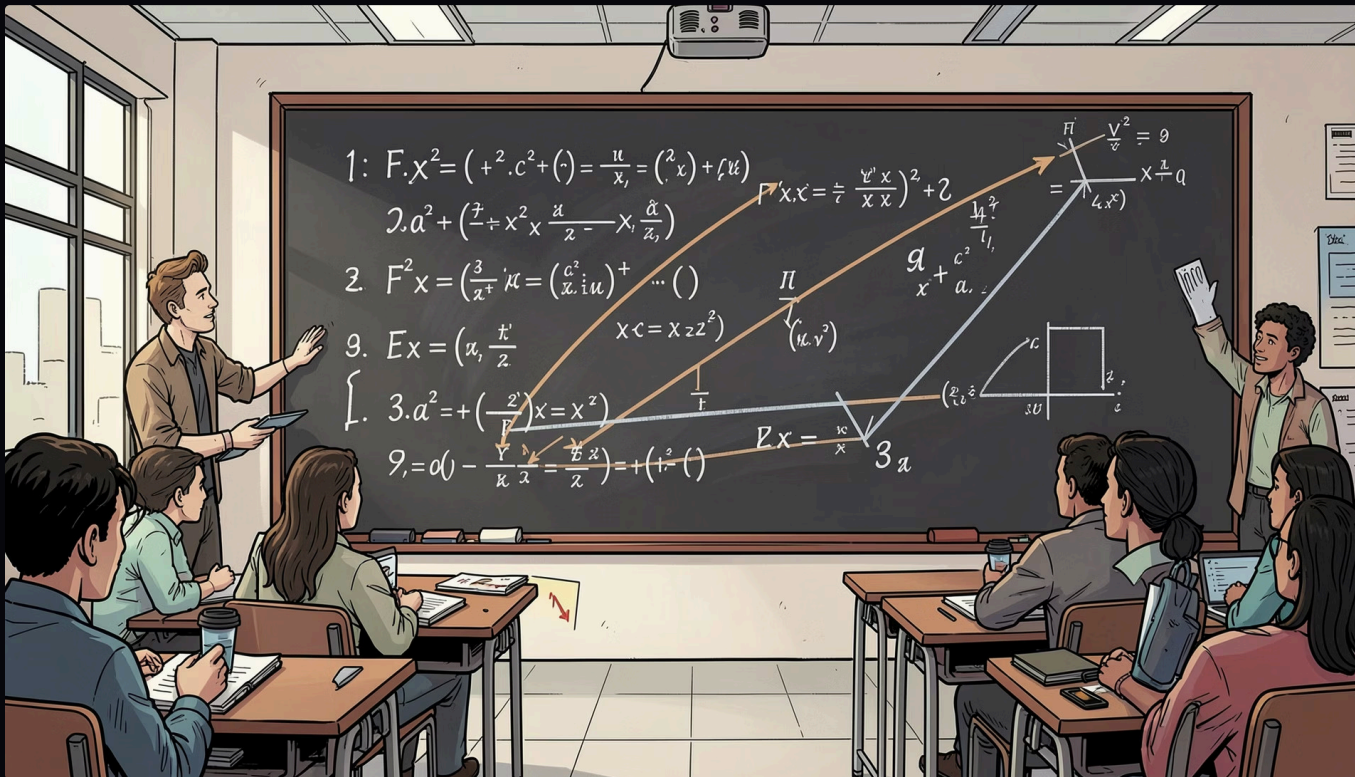
$T = 2u \sin\theta / g$
Total time in air

Maximum Range

$R = u^2 \sin 2\theta / g$
Max at $\theta = 45^\circ$



Projectile Motion — Key Formulas



Maximum Height

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

Horizontal Range

$$R = \frac{u^2 \sin 2\theta}{g}$$

Equation of Path

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta}$$

Velocity at Any Point

$$v = \sqrt{v_x^2 + v_y^2}$$

Uniform Circular Motion

What is UCM?

Motion along a circular path with **constant speed**. Direction changes continuously, so velocity is NOT constant — there is always acceleration.

Centripetal Acceleration

$$a_c = v^2/r = \omega^2 r$$

Always directed **towards the centre** of the circle.

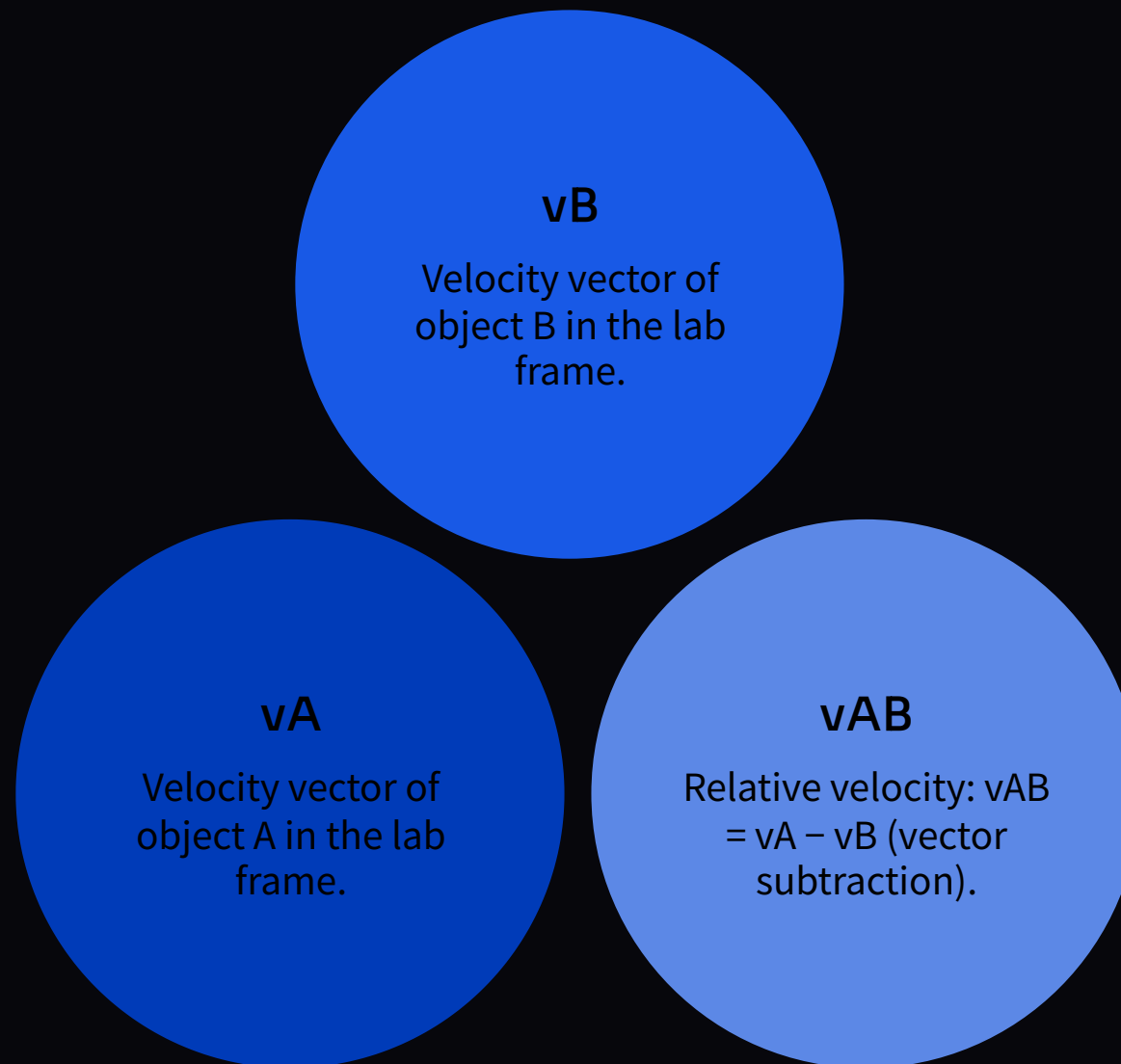
Angular Quantities

- **Angular displacement:** θ (radians)
- **Angular velocity:** $\omega = d\theta/dt$
- **Angular acceleration:** $\alpha = d\omega/dt$
- **Relation:** $v = r\omega$

✓ Frequency $f = 1/T$ and $\omega = 2\pi f$

Relative Velocity in 2D

Velocity of object A **relative to** object B is the velocity of A as observed from B's frame of reference.



The relative velocity is found by vector subtraction of the two velocity vectors.

Formula

$$\mathbf{v}_{AB} = \mathbf{v}_A - \mathbf{v}_B$$

Magnitude: $|\mathbf{v}_{AB}| = \sqrt{v_A^2 + v_B^2 - 2v_A v_B \cos\theta}$

Applications

- Rain falling on a moving person
- Swimmer crossing a river
- Aircraft navigation with wind



REVISION

Chapter 4 — Quick Revision



Vectors

Magnitude + Direction. Use $\hat{i}, \hat{j}$ notation for components.



Projectile

Parabolic path. Max range at 45° . Treat x and y independently.



Circular Motion

$a_c = v^2/r$ towards centre. $v = r\omega$. Speed constant, velocity changes.



Relative Velocity

$v_{AB} = v_A - v_B$. Essential for rain, river, and navigation problems.